Imperial College London

Lecture 5

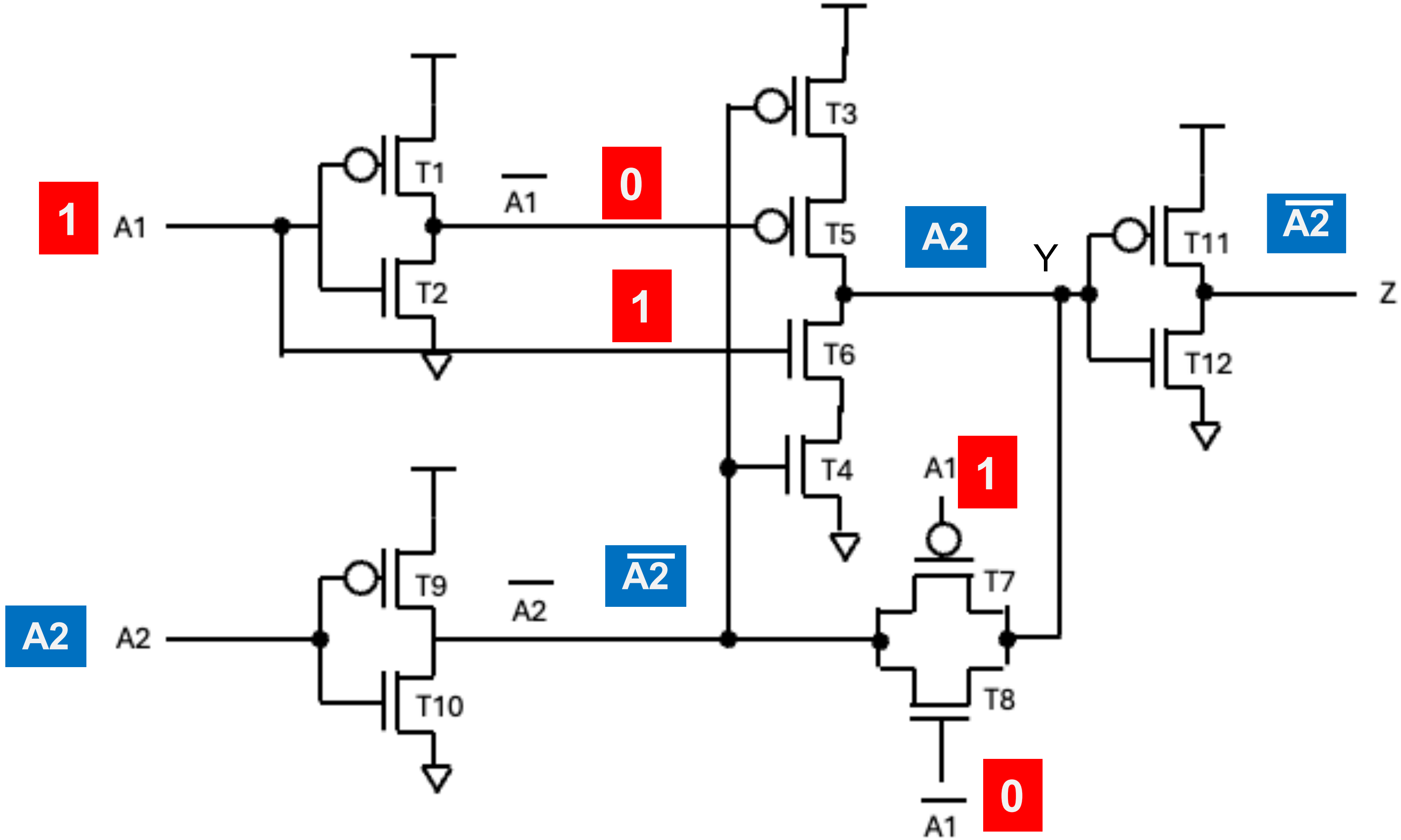
Combinational Circuits

Prof Peter YK Cheung

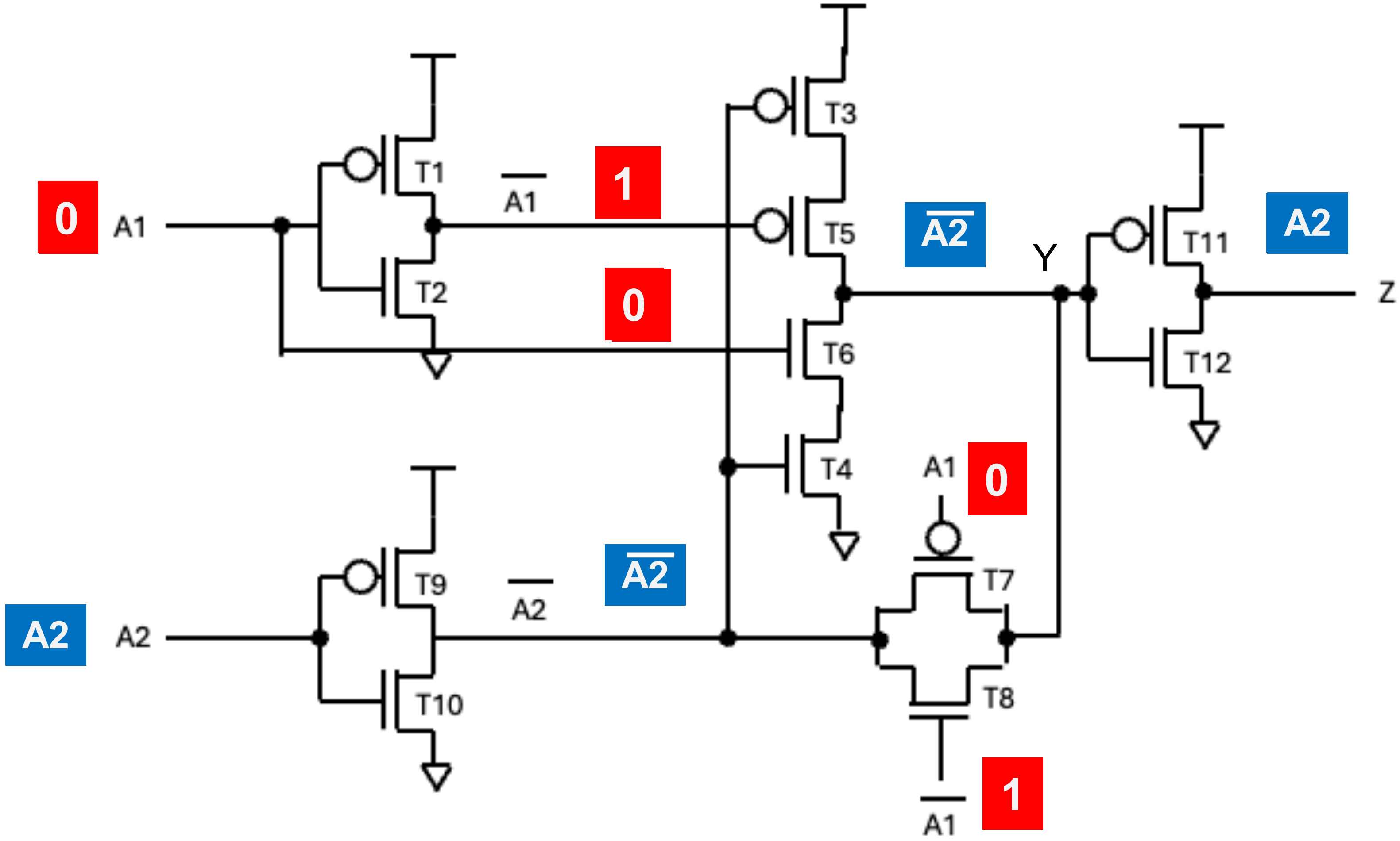
Department of Electrical & Electronic Engineering

URL: www.ee.ic.ac.uk/pcheung/teaching/EE4-VLSI/
E-mail: p.cheung@imperial.ac.uk

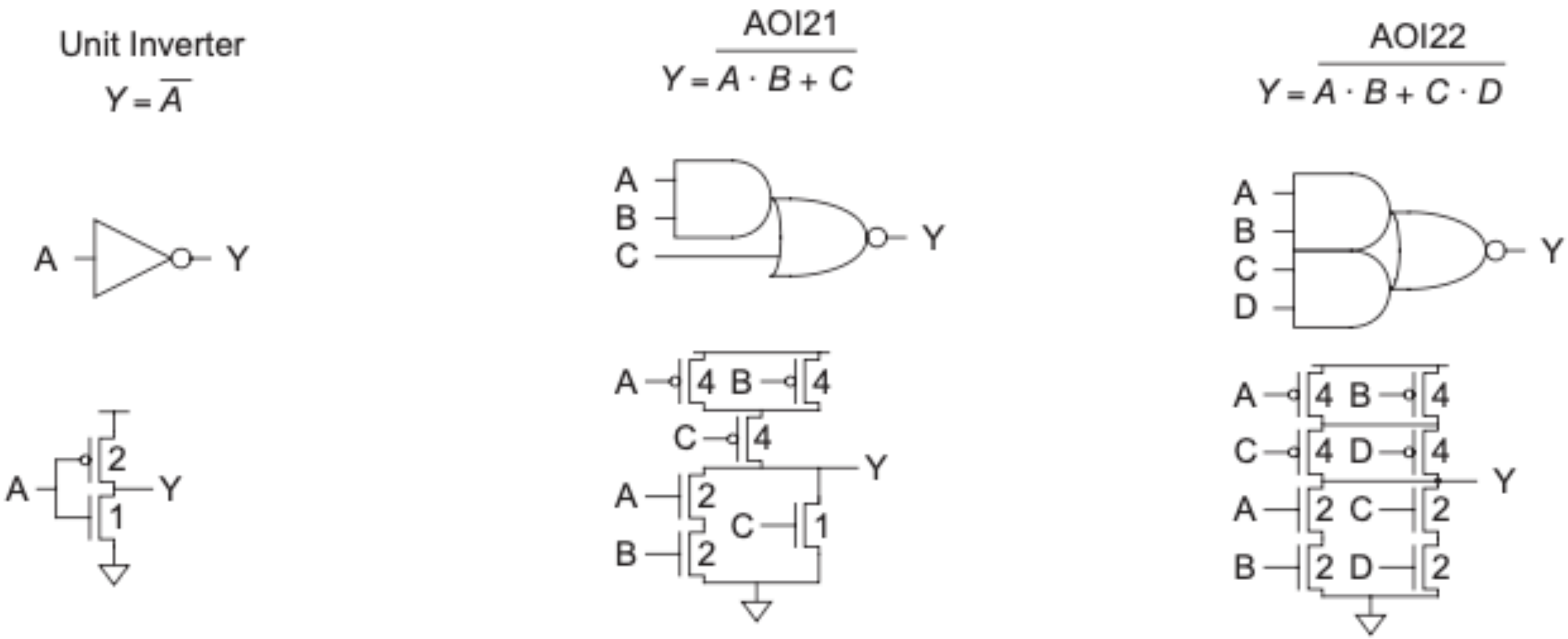
XOR gate explained (1)



XOR gate explained (2)



Compound Gates

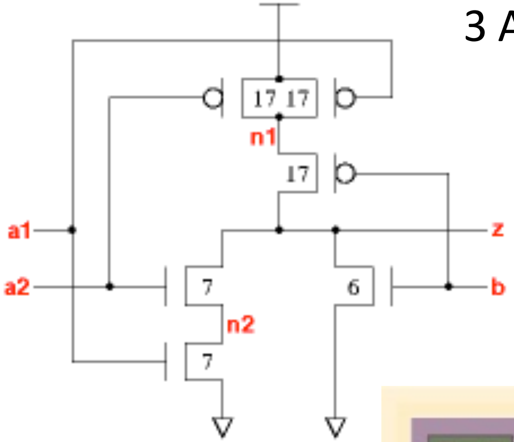


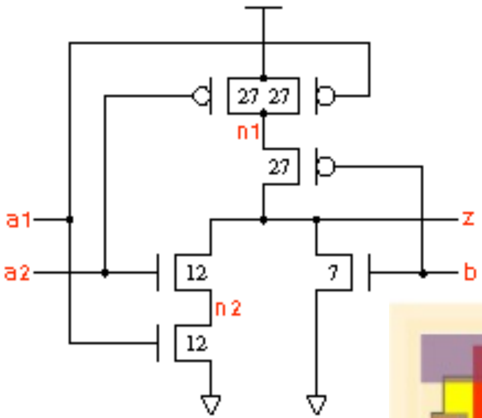
W&H p330

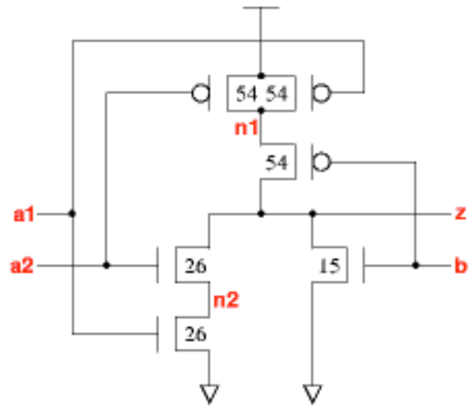
vsclib Standard Cell Library – aoi21

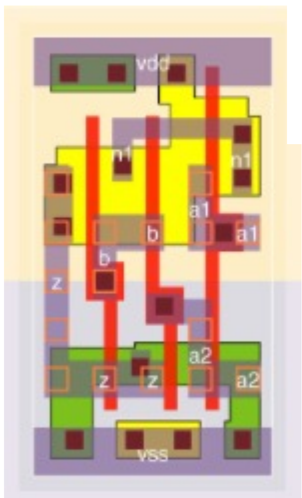

```
z:
((a1*a2)+b)'
```

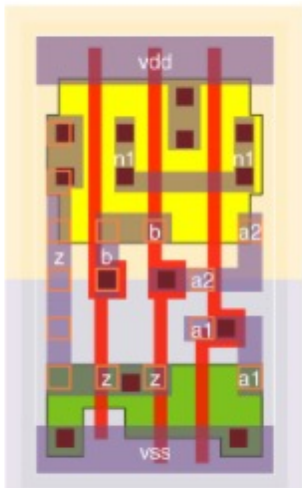
3 AOI31 cells with different drive strengths.

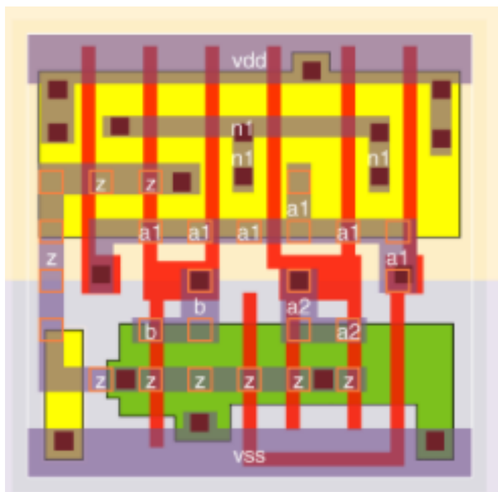




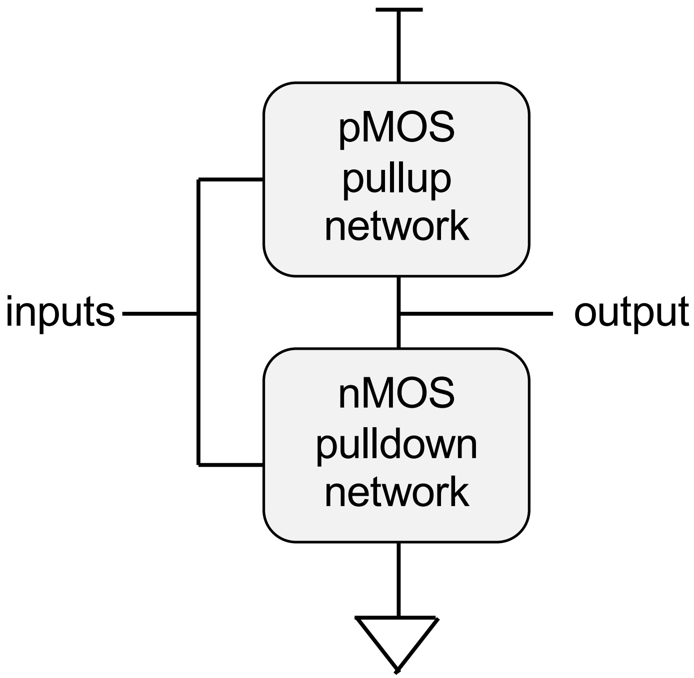


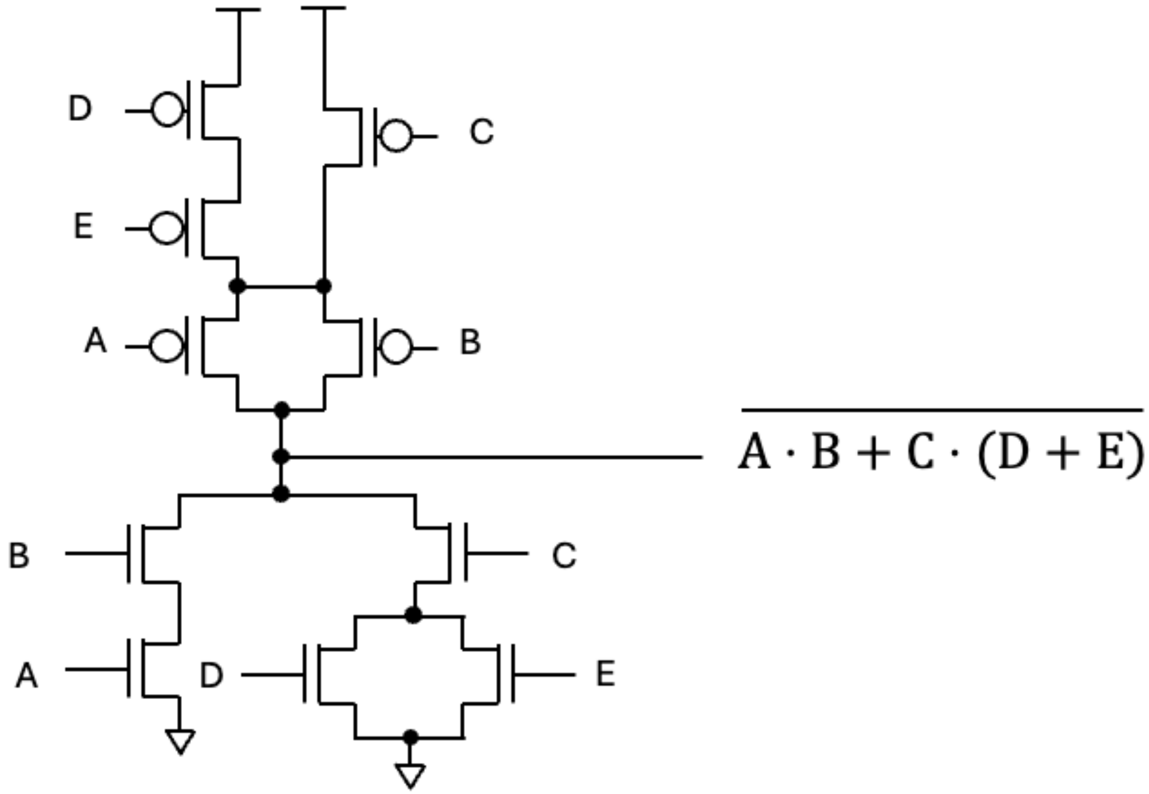




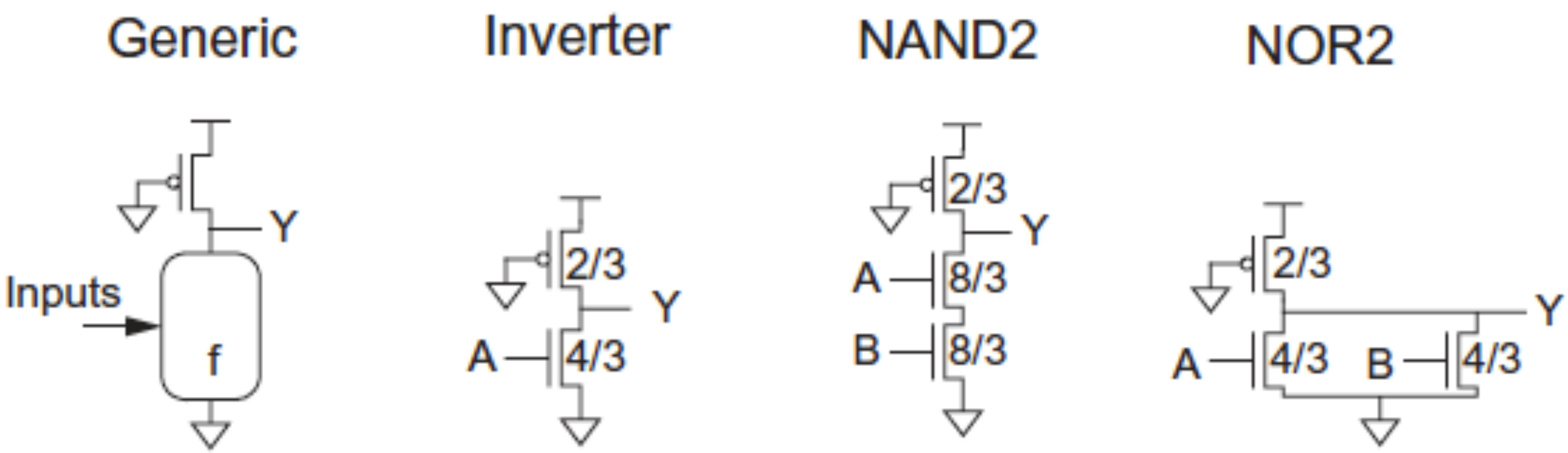


Complex Compound Gate



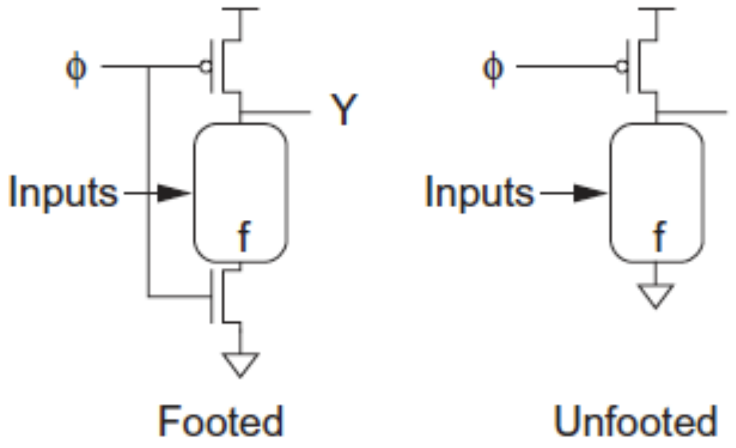


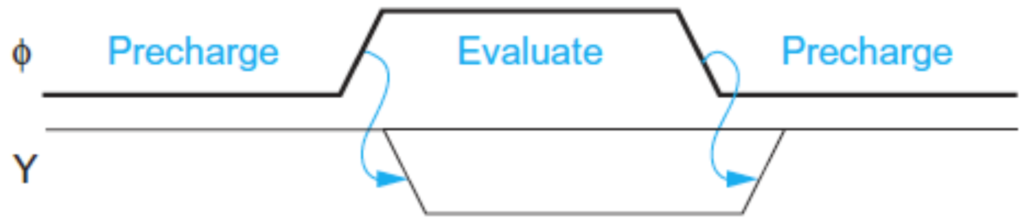
Pseudo-nMOS Logic Gates

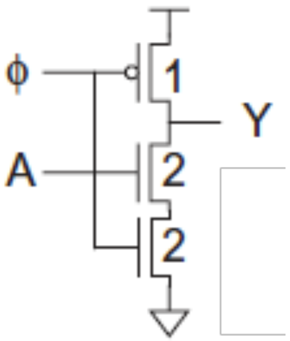


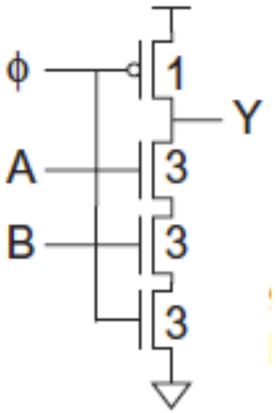
W&H p335-336

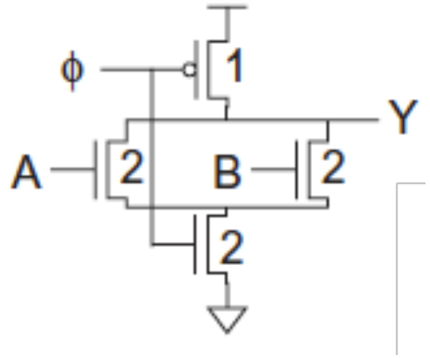
Dynamic Logic Gates





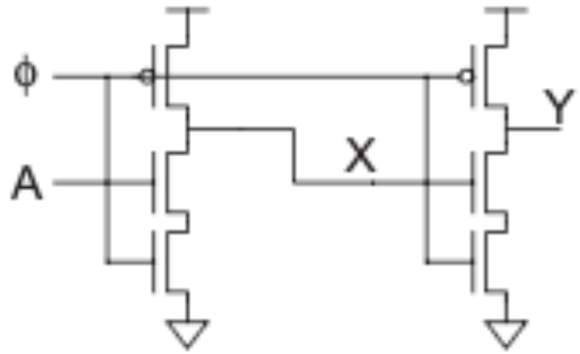


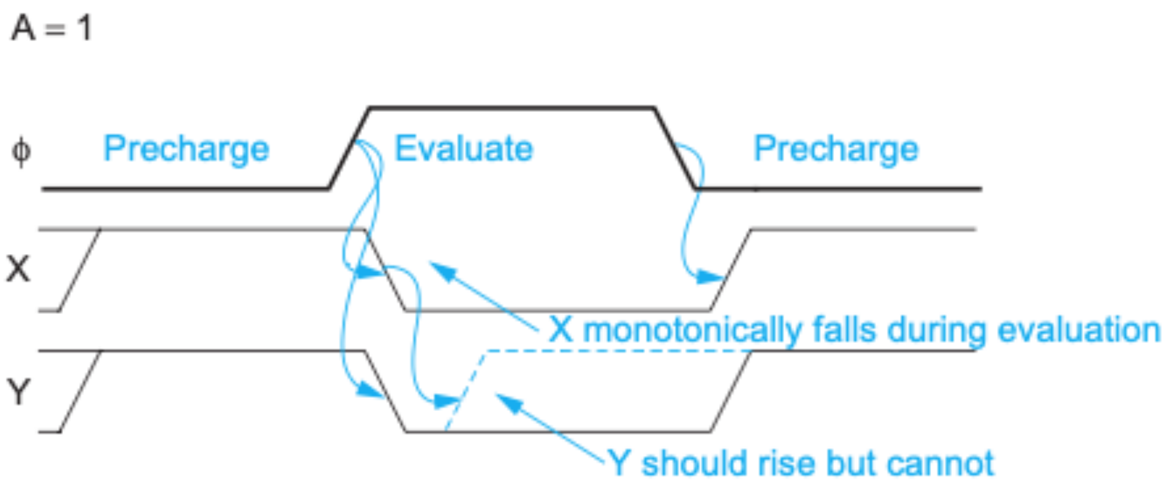




W&H p339-341

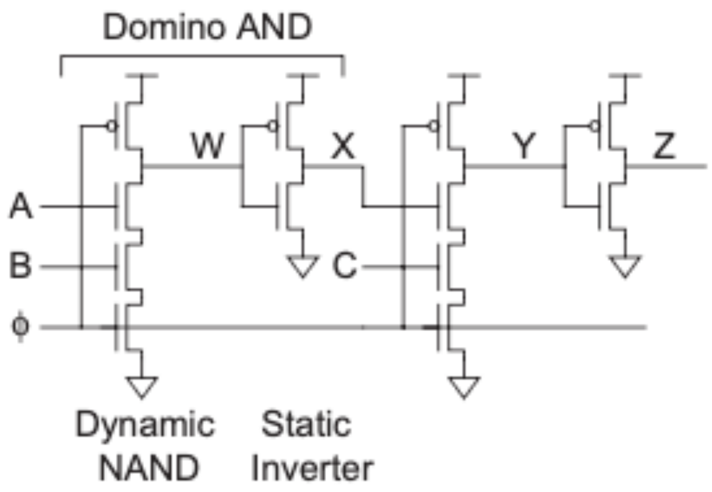
Problem with Dynamic Logic Gates

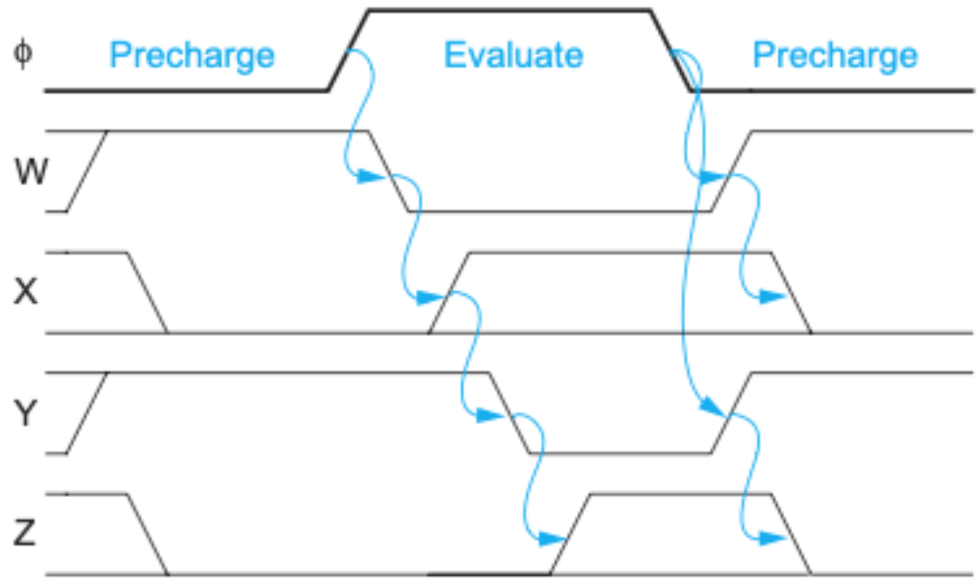


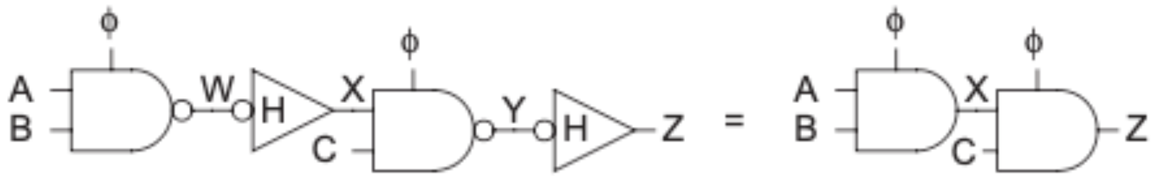


W&H p341

Domino Logic

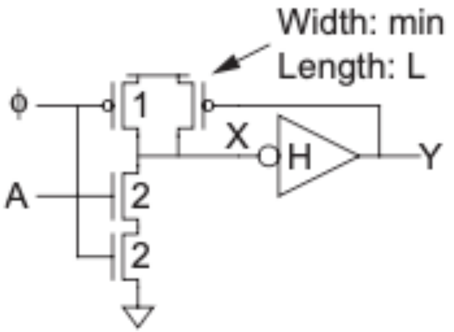


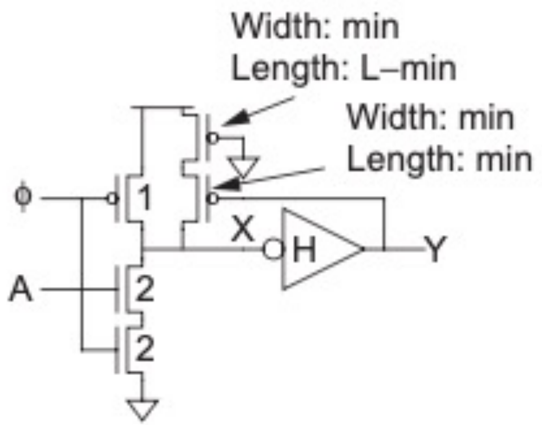


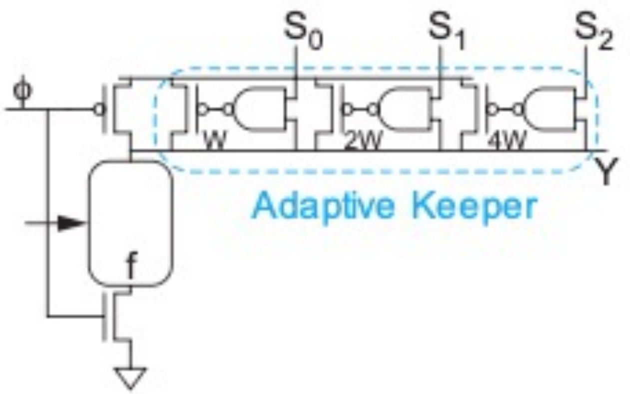


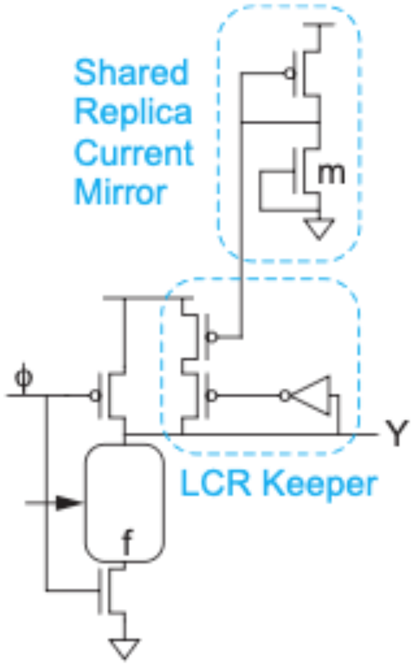
W&H p341-342

Keepers for Dynamic Logic



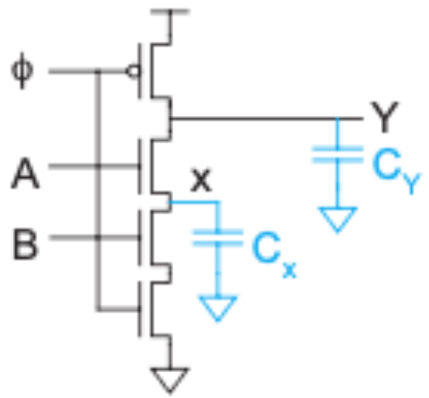


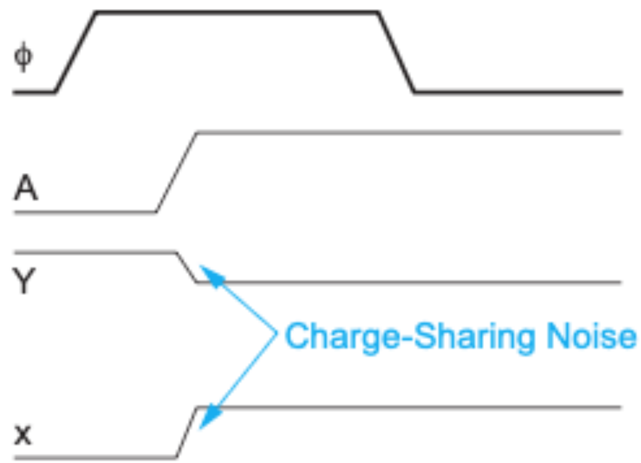


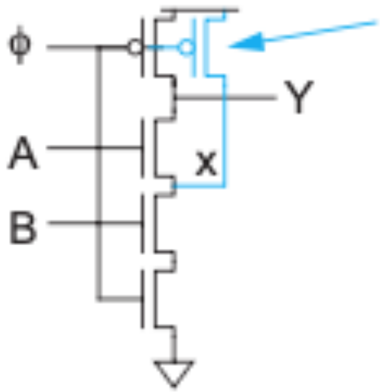


W&H p343-344

Charge Sharing in Dynamic logic







W&H p345